MIS 201 · WEEK 8 · MONDAY 11/9

Ch. 8: Ethics & War
Holding two things

Part 1. The old rules of war, and what AI does to each one.

Part 2. Ukraine's drones, the future of war, and the rules you'd pick.

Wednesday 11/11: no class. Veterans Day.

This week: one class, two things due Friday

DUE FRIDAY

WHAT

Lab 8 writeup and the Chapter 8 quiz.

WHEN

Friday 11/13, 11:59 p.m., on Canvas.

WEDNESDAY 11/11

No class. Veterans Day.

LAB 8: AT HOME

The whole lab is done at home this week. Plan on about two hours.

Hang on to your answer to today's question 5. The lab asks for it again.

Watch: who decides to pull the trigger?

DW · 2.5 MIN

DW NEWS · JUNE 3, 2026

Ukraine Is Now the World's AI War Lab

An engineer, an analyst, and an official on where people fit when AI speeds up every step.

https://www.youtube.com/watch?v=Gfqdf4JFErU

WATCH FOR

1. Where does AI show up between spotting a target and striking it?
2. What does the analyst actually do when he "confirms"?
3. Recommending vs. doing: where is the line?

On the board: who decides to pull the trigger?

PART 1 · CHAPTER 8 · ABOUT 50 MINUTES

AI and War: Ethics

This week we hold two things at once: respect for the people who serve, and hard questions about the tools they're handed.

Five ethics questions for AI in war

THE ETHICS QUESTION	THE OLD RULE	WHAT AI CHANGES
Who may be harmed?	Distinction: only attack soldiers	**A model scores who's a soldier, and someone picks the cutoff**
Who decides?	A person stays responsible for force	**A person still approves each strike, but with only seconds to do it**
Who do we trust?	Human judgment	**Trust shifts to a machine**
Who's responsible for the tool?	The builder answers for its use	**Built for one use, sold for another. Who answers?**
Who answers when it fails?	Accountability	**More decisions, less clear blame, still investigated in secret**

The kill chain: AI helps, people decide

WHO DECIDES?

1. FIND	2. FIX	3. TRACK	4. TARGET	5. ENGAGE	6. ASSESS
sort through satellite and drone images	confirm the target	follow target or location	match to a weapon	the strike	did it work?
Predictive AI	*Predictive AI*	*Predictive AI*	*predictive + generative*	*Embodied AI*	*Predictive AI*
People: analysts check the results	**People:** analysts confirm the results	**People:** analysts keep watch on any updates	**People:** government, officers and lawyers check	**People:** commanders approve	**People:** people judge the result

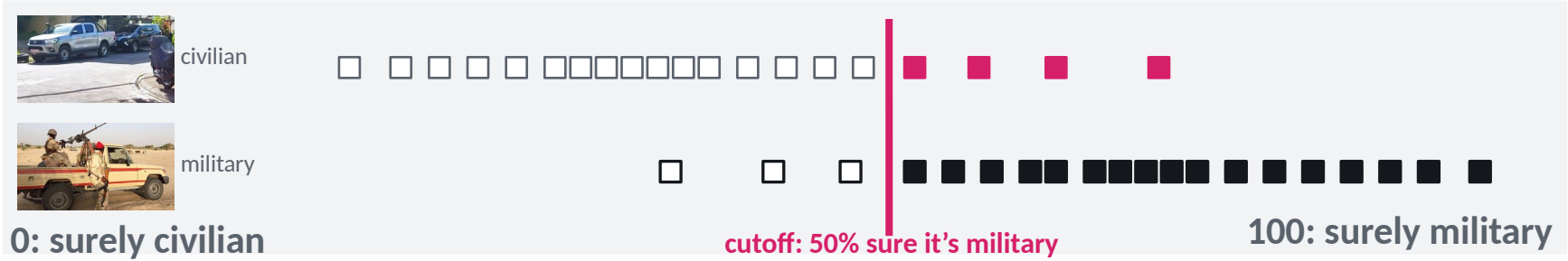
AI came in from the left, in the unglamorous steps. It has been moving right.

A person approves at every step. But every step AI reaches **shortens the time that person has to say no.**

Distinction: a moral rule, and a setting someone picks

Distinction = the oldest rule of war: tell soldiers from civilians, and only attack soldiers.

How AI does it: an image-recognition model scores each vehicle from 0 to 100. It's a **classifier**, like Chapter 6's résumé screener. Someone picks the cutoff, called the threshold.



4 CIVILIANS FLAGGED

Pink squares. The cost lands on civilians.

3 MILITARY MISSED

Hollow squares. The cost lands on soldiers.

Someone chooses the cutoff

Invented scores. Photos: Ethan Llamas, CC BY-SA 4.0; Nicolas Pinault/VOA, public domain.

Meaningful human control: what would it take?

WHO DECIDES?



A Ukrainian drone pilot. Who is in the loop today.

Photo: General Staff of the Armed Forces of Ukraine, CC BY 4.0.

Three conditions for human control to mean anything:	UNDERSTANDING knows why the machine proposed this	TIME enough to think, not just react	AUTHORITY can actually say no, and it sticks
Pilot, reviewing a route	yes	yes: minutes	yes
Soldier approving machine targets	**a label, not a reason**	**seconds**	**unclear**

Perhaps in war, a **human in the loop is not enough.**
Ask: **do the humans understand, do they have time, and can they say no?**

Automation bias: when the human trusts the machine

WHO DO WE TRUST?

Automation bias = trusting a usually-right machine without checking, especially under time pressure.

PETROV · USSR, 1983	
Machine said	5 U.S. missiles launched
Time	minutes
Trained to	report it up the chain
He	doubted it
Result	a false alarm, caught

PATRIOT CREWS · IRAQ, 2003	
Machine said	enemy missile incoming
Time	seconds
Trained to	trust the system
They	approved it, twice
Result	2 friendly jets downed, 3 aircrew killed



A Patriot battery (Turkey). Photo: U.S. DoD, Glenn Fawcett, public domain.

From the DW clip: the analyst confirms what the machine claims. How carefully, at this speed?

Same setup, a person between a computer and a weapon. **A human in the loop only helps if the loop lets them doubt.**

Dual use: who draws the line?

WHO'S RESPONSIBLE?

Dual use = the same technology can serve civilian and military purposes.

GPS was built for the military and now finds your pizza. The AI that writes a marketing email can also write an intelligence summary.



The ethics question: if you build it, are you responsible for how a customer uses it?

GOOGLE · 2018

Employees protested a Pentagon contract to analyze drone footage. Google let the contract lapse.

ANTHROPIC · 2026

Refused two military uses. The Pentagon pushed back, and it's in court.

Who should draw the line: the workers, the company, the customer, or the law?

GPS satellite: NASA, public domain.

Who answers when it fails?

WHO ANSWERS?

THE OLD WAY

An officer orders a strike.

↓

A soldier carries it out.

↓

If it's wrong, they answer for it.

WITH AI IN THE LOOP, THE BLAME SPREADS OUT

- The company that built the model
- The old data it learned from
- The analyst who approved it in seconds
- The commander who trusted it
- The machine, after the lock

Each has a piece. None has all of it.

Someone has to answer. AI makes it harder to say who, and the side that made the mistake usually investigates itself, in secret.

Minab: what we know, and how we would know

AS OF NOV 2026

WIDELY REPORTED

Feb 28, 2026: a U.S. missile hit a girls' elementary school in Minab, Iran. More than 100 children were killed. Early Pentagon findings: a targeting mistake. The school had once been part of the base next door.

REPORTED BY UNNAMED OFFICIALS

The Pentagon's review found old intelligence, old satellite images, and too much trust in a Palantir AI targeting system. Some staff expected the system to catch the stale data. It didn't. The report hasn't been released.

NOT ESTABLISHED

That AI chose the target.

That Claude or any chatbot was involved.

The U.S. hasn't released its findings or publicly accepted responsibility.

Sources in the speaker notes. Check for updates before class.

PART 2 · THE CASE · ABOUT 50 MINUTES

Need a break? Go for it.
Ukraine's drones, and the future of war

AI is changing what war is: who fights, who decides, and who answers.

Then the question for next week in Geneva: what should the rules be?

Why this case?

CASE

WHAT'S HAPPENING

Since Russia's 2022 invasion, Ukraine has fought with millions of cheap drones, flown by soldiers in video goggles.

THE PROBLEM

Russia jams the radio link, so the drone goes blind in the last stretch before the target.

THE FIX

A small AI unit that keeps steering after the link drops. A person picks the target. The machine finishes.



Why it matters in November 2026: AI drones are in use on both sides right now. Ukraine's president has warned AI could be leading battlefield decisions as soon as next year. And next week in Geneva, countries decide whether to start writing rules.

This is a country defending itself. The question isn't whether Ukraine should defend itself. It's what happens to human control when the fighting gets this fast.

Photo: ArmyInform, CC BY 4.0.

How the drone's AI takes over: the lock

WHO DECIDES?

Radio drone with terminal guidance

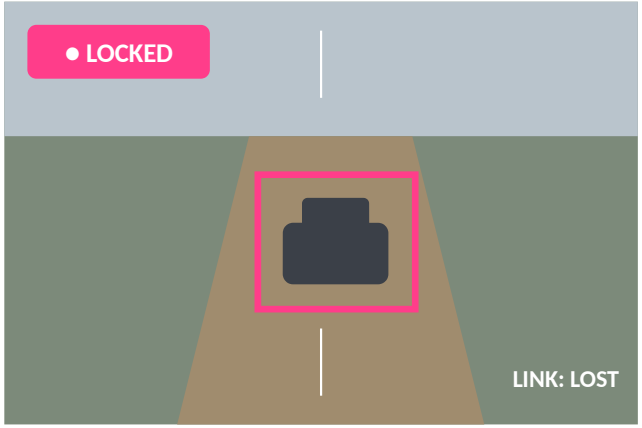
pilot flies it over a video link | jammed: AI holds the box

the lock: the last moment a person can call it off

Fiber-optic drone

pilot flies it over a thin cable, all the way to impact

Costs: shorter range, slower flight, a cable that can snag.



What the pilot sees: box the target, and the drone holds it after the link drops.

How it works: the pilot flies the drone and draws a box around the target. That's the lock. If jamming cuts the radio link, a small AI on the drone keeps steering toward whatever is in the box. A person picks the target; the machine finishes the job.

Illustration, not real footage.

Watch/discuss: the future of war

60 MINUTES · 12 MIN

60 MINUTES · HOLLY WILLIAMS

Inside Ukraine's drone war

Ukraine's drone war: robots that take the risk, a new version every week, and what comes next.

https://www.youtube.com/watch?v=jgT8URb0HO8

WATCH FOR

1. Where is the human in each drone: flying it, approving it, or gone?
2. "Not risk a human life": who does a robot protect, and who does it not?



Hold this question: ten years from now, where is the human in the loop?

Ground robot photo: ArmyInform, CC BY 4.0.

AI is changing what war is

THE BIG QUESTIONS

WHO FIGHTS	Drones and robots, flown from bunkers far away. Drones cause an estimated 80% of combat casualties in Ukraine.
WHO DECIDES	Machines find targets and recommend strikes. People get seconds to say yes.
HOW FAST	A new drone version about every week. Machines decide in microseconds; people take minutes.
WHO ANSWERS	Many hands, and no one holds all the blame.

QUESTIONS NO ONE HAS SETTLED

Should a machine ever decide to take a life?

How much human control is enough?

Who answers when it goes wrong?

This changes what humanity has meant by war, and how people take part in it. Next week in Geneva, countries decide whether to write the rules.

Who makes the rules? A UN decision next week

WHO ANSWERS?

2014

UN talks on autonomous weapons begin in Geneva.

SEPT 2026

128 countries agree on draft elements. Not a treaty. Not binding.

OCT 29, 2026

163 countries vote in a UN committee to start treaty talks. 5 vote no.

NOV 16 TO 20

Countries decide whether to open formal negotiations.

THE CATCH

These talks run by consensus. One country can block a treaty, however many want it.

WHO RESISTS

The U.S., Russia, and others. They argue the existing laws of war already cover it.

If nobody outside finds out when it fails, **the only check left is a rule written in advance.**

What should the rules be?

YOU DECIDE

BAN

Weapons that pick their own targets.

Cost: Rivals may not sign, and drones get built anyway.

REQUIRE CONTROL

A person with understanding, time, and authority for every strike.

Cost: Slower, and who checks that it was real?

PROTECT PEOPLE

No AI weapons that target people. Vehicles and objects only.

Cost: The line blurs: people ride in vehicles.

LOG EVERYTHING

Every AI-assisted strike recorded and reviewed by someone outside.

Cost: Secrecy. Militaries will push back.

NO NEW TREATY

The existing laws of war already apply.

Cost: "Appropriate human judgment" stays undefined.

Discuss: our five questions, and the future of war

GROUP DISCUSSION

1. **Is a human in the loop enough?** In war, a person may approve each strike with seconds and only a label to go on. Does "human in the loop" go far enough? What would enough look like?

2. **What counts as war now?** AI now finds targets, steers drones, and suggests decisions. As machines make more of the choices, how does that change what we have always thought of as war, and who takes part in it?

3. **Who answers?** When an AI-assisted strike goes wrong, or an autonomous weapon acts on its own, who should be held accountable: the soldier, the commander, the company, or the country?

4. **Pick your Geneva rule.** You're a delegate in Geneva next week. Pick one of the five rules from the last slide. What does it cost, and who pays: soldiers, civilians, or the countries that follow the rules?

5. **The future of war:** Picture a war ten years from now, with swarms. Is the human flying each drone, approving each strike, or just setting the rules before launch? Which future worries you most?

1. Groups, 7 min.

Answer two: the one I assign (1 to 4), plus #5. Pick a leader.

2. Whole class, 5 min.

One leader per question. Then everyone on #5.

3. Vote.

Hands up for your Geneva rule from #4. Name its cost.

Read your answer aloud as if a veteran were in the room.

BEFORE YOU GO

Attendance before 4 p.m., please!

Upload your notes to Canvas right after class.

Wednesday 11/11: no class. Veterans Day. Lab 8 is done at home this week.

Friday 11/13, 11:59 p.m.: Lab 8 writeup and Chapter 8 quiz.

Want a head start? Stay and start Lab 8 with a partner.